

NEMATOCYSTS OF *MICROSTOMA*.

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For a long time the nematocysts of *Microstoma* were considered to be structures that the worm elaborated as did *Hydra*. Martin ('08) published an account of the manner in which *Hydra* is attacked by *Microstoma* and indicated his conviction that the nematocysts of *Microstoma* were derived from the *Hydras* that had been eaten. Kepner (11) published an account of the histological details involved in the process of transporting the nematocysts of *Hydra* from the enteron to the epidermis of *Microstoma*. In this account the inference was made that this complicated series of phenomena must mean that *Microstoma* handled these nematocysts in this manner in order to use them. This inference was early challenged by Glaser (11). A tentative reply was given Glaser's note in *Science*, Volume 34, page 213, 1911.

The details of the processes involved when *Microstoma* deals with the nematocysts are so variable and so intricate that it was felt that these processes could mean only one of two things, namely, either this was a method of eliminating indigestible foreign bodies or a method of securing weapons of defense or perhaps offense.

From time to time since 1911 efforts have been made to determine which of the two meanings was behind this conduct of *Microstoma*. This paper represents the results of our efforts that have thus extended over a decade and have been taken up each year of this decade to a certain extent.

MATERIAL AND METHODS.

Microstoma caudatum is found rather abundantly in the early spring months in the submerged detritus along the banks of fresh water ponds about the University of Virginia. Sometimes *Hydras* are found living with it in great numbers; at other times

there are few Hydras. When many Hydras are present it is difficult to find *Microstomas* that lack nematocysts. When there are few Hydras living with *Microstoma* in nature, specimens that lack nematocysts are readily obtained.

It has been found that by placing *Microstoma* in watch-glasses containing fresh water and some fragments of water soaked, dead leaves they can be kept indefinitely; provided the air be kept free of coal gas, illuminating gases, and formaldehyde vapor.

NEMATOCYSTS PASSING FROM ENDODERM INTO MESODERM.

While this paper is not dealing with the methods by which *Microstoma* handles nematocysts, it is of interest to record that Kepner and Whitlock in 1917 observed a nematocyst being carried down through the body of an endodermal cell and delivered through the basement membrane into the mesenchyme. This was a very slow process. The movement of the nematocyst could be appreciated only when its position was recorded and then five minutes later its position again observed. Its rate was not accelerated as it passed through the basement membrane into the mesoderm. The most remarkable feature of this was that the obtuse end was directed towards the path along which the nematocyst was being transported.

NEMATOCYSTS UNIFORMLY DISTRIBUTED.

Attention might be called to the additional fact that when these nematocysts are taken up by the mesenchymal cnidophages they are distributed uniformly at the epidermis by these attending cells. So that immediately after a *Microstoma* has acquired a supply of nematocysts at its surface, these nematocysts are uniformly distributed. It does not appear, however, that if the anterior end, or any other region, lose its quota of nematocysts that some will be taken from other regions of the body to take the place of those lost. In this manner specimens sometimes come to have a uniform distribution of nematocysts except for some region (e. g. "head") that is free from them. Despite this fact, the subtle manner in which the cnidophages coöperate to bring about a uniform distribution of the foreign nemato-

cysts is remarkable and lends weight to the inference that the nematocysts are being collected by the *Microstoma* to be used.

NEMATOCYSTS RETAINED INDEFINITELY.

Again if the nematocysts of *Hydra* are carried out to the surface of *Microstoma* by way of eliminating them as indigestible foreign bodies, the question arises as to why they are held so long at the surface. Within twelve hours after a *Hydra* has been ingested by *Microstoma*, the nematocysts are distributed over the rhabdocoele's body. The process of getting these objects to the surface of the flatworm is, therefore, a matter of about twelve hours. When, however, they arrive at the surface they are held there indefinitely. Even when animals are kept under adverse conditions in the laboratory and they show more and more conspicuous decline, the nematocysts are yet retained. Specimens have been seen growing weak and beginning to rupture or break up and yet retain the nematocysts over the epidermis that remained intact. This retention of the nematocysts, likewise, suggests that they have been collected for use.

AN EXPENSIVE METHOD.

Again this is an expensive method of dealing with objects that could be thrown out at the mouth as are other indigestible materials. For in this process much energy must be spent by the endodermal cells and cnidophages in handling the nematocysts; and, further, when the nematocysts are discharged they carry with them their attending cnidophages. The cnidophages, therefore, in attending the nematocysts act against their own welfare in that in the end they lose their own lives.

MICROSTOMA INCURS A DANGER.

Even the *Microstoma* as a whole feeds upon *Hydra* at a risk. For sometimes *Hydra* turns upon *Microstoma* and eats it. *Hydra*, however, seems to have difficulty in ingesting *Microstoma* as the following observation of January 18, 1917, shows. The *Microstoma* at once played along the side of *Hydra viridis* that had been placed with it. The worm passed to and fro along the surface of the polyp's body. Twice it came to rest amidst

the bases of the tentacles. The third time it came to lie over the mouth of the *Hydra*, the latter began an effort to ingest the *Microstoma* (Fig. 1). First the *Hydra's* peristome opened up against the ventral side. This region of the peristome did not fix the *Hydra* to the *Microstoma*, for the mouth of *Hydra* glided posteriorly along the ventral side of *Microstoma* and around its posterior end as indicated by the arrow in Fig. 2. When the widely expanded peristome of *Hydra* had come to be applied over a great part of the dorsal and lateral surfaces of the *Microstoma*, the *Hydra* pressed the latter between its peristomal bell and its body and appeared to have the worm in a very serious position (Fig. 3). But from this embrace the *Microstoma* soon glided. As it escaped it showed a ruptured region of its body. This wound healed in a little while and then *Microstoma* went back and played along the surfaces of *Hydra's* body. The next time it came into the tentacular zone of *Hydra*, the polyp succeeded in grasping the *Microstoma* head-on and forthwith ingested it (Fig. 4). Since, therefore, *Microstoma* incurs a greater danger in seeking out *Hydra* than it does when it feeds upon small annelids, crustacea and other small animals and plants and since, further, it does not appear to seek *Hydra* primarily for food as will be shown later, the inference is strong that it is seeking *Hydra* for some unique end.

Microstoma ABLE TO DRAW NEMATOCYSTS FROM ITS BODY
WHEN WOUNDED BY *Hydra*.

Moreover, *Microstoma* seems to have developed an adaptive secretion by its epidermal glands that may be taken to be either rhabditic glands or the homologues of such glands which are found in other rhabdocoeles. The following observations indicate the adaptive functioning of these glands. February 15, 1917, a *Microstoma* was placed with a small *Hydra fusca*. The *Microstoma* began playing about the five tentacles. Immediately the *Hydra* firmly grasped it between its tentacles (Fig. 5). The *Microstoma* escaped from the *Hydra's* embrace. As it swam clear of the polyp, two mucous masses were to be seen, one at the posterior end and one on the right side. (Fig. 6). The lateral mass grew and as it grew drew the two enclosed nemato-

cysts out from their anchorage within the worm's body and then glided along the right side of the body posteriorly to where it fused with the posterior mass, which had three nematocysts enclosed within it. This compound mass of mucus, now containing five freed nematocysts, was sloughed off from the *Microstoma's* body and left behind. After that the *Microstoma* seemed to be enveloped in a transparent mucus sheath to which small dead objects adhered and were dragged about with the *Microstoma* until they became entangled with the tentacles of the *Hydra*. These tentacles stripped the foreign bodies from the *Microstoma's* adhesive surface. Another observation of this sort was made January 29, 1917. A *Microstoma* was placed with a large budding *Hydra fusca*. It at once attacked the *Hydra*. The *Hydra* discharged a nematocyst into the anterior end of *Microstoma*. The *Microstoma* contracted so vigorously as to break apart the two nearly formed zooids that had been developing through fission. About the deeply anchored nematocyst the anterior zooid now secreted a local mass of mucus. This mass of mucus grew until it formed a column whose length was more than two-thirds that of the zooid. Next the posterior zooid encountered the bud of the *Hydra*. As it passed the *Hydra*, the latter stung it in its posterior end along the left side. About this nematocyst a local secretion of mucus appeared and as the mass of mucus grew in length the long stinging thread of the nematocyst was dragged from the *Microstoma's* body as had a mucous column dragged the nematocyst from the body of the anterior zooid. Eventually the mucous masses were discarded and the two zooids appeared to be no worse for their experience.

All the above shows an intimate relation existing between *Microstoma* and *Hydra* which is peculiar and must have some meaning while much of the above indicates that the meaning of this conduct centers about the nematocysts of *Hydra*.

Microstoma WITH FEW NEMATOCYSTS REACTS TO *Hydra*
READILY.

Moreover, a *Microstoma* that has few or no nematocysts behaves differently towards *Hydra* than does one that has a com-

plete quota of nematocysts at its surface. On December 8, 1916, a *Microstoma*, that contained seven or eight nematocysts and so greatly filled with food that the pharynx of its posterior zooid was everted, was placed with a *Hydra*. Within 12 minutes the *Microstoma* had egested some of its food and had torn off a part of the *Hydra's* oral end. Two days later a second well-fed specimen, that contained but two nematocysts, was placed with a *Hydra*. This specimen ate the *Hydra* within twenty minutes. Table I., shows that of 42 specimens containing none or few nematocysts, 6 (within two minutes after being placed with Hydras) set to work trying to feed upon the polyps, but five of them got severely wounded while the sixth was eaten by the *Hydra*. Two others were eaten by the *Hydra* within twenty-four hours. Of the remaining 34 specimens, 9 had eaten Hydras in periods ranging from 2 minutes to 20 minutes; while 20 accepted Hydras in periods ranging from 24 hours to 1 hour; and 3 remained with *Hydra* either 5 or 2 days before they accepted *Hydra*. Two specimens remained with *Hydra* 24 hours and a third 4 days when they were lost or had died without having accepted *Hydra*. The conspicuous feature of Table I. is that in many cases *Hydra* was accepted within a period of minutes, many others within a period of hours, while only a few were accepted within a period of days.

Microstoma WITH MANY NEMATOCYSTS REACTS TO *Hydra*
TARDILY.

Table II. makes a sharp contrast with Table I. in this respect. For this table shows the reactions of 18 *Microstomas*, that had either many or a full quota of nematocysts. All, except specimen 15, were kept from food for a day or more and yet only one of them, specimen 7, reacted to *Hydra* within 24 hours and this one contained only about one-third of a full quota of nematocysts. Specimen 8 remained with *Hydra* 24 hours when it was lost, without having accepted *Hydra*. Except for these two specimens, all the others reacted to *Hydra* not within periods of hours or minutes but of days and that despite the fact that they had no food for at least a day. Specimens 17 and 18 were kept away from food in each case four days. They lived four

TABLE I.
NEMATOCYSTS OF *Microstoma* NONE OR FEW.

Specimen.	Number of Nematocysts.	Time with <i>Hydra</i> .	<i>Hydra</i> Eaten.	<i>Hydra</i> Not Eaten.	<i>Microstoma</i> Eaten.	<i>Microstoma</i> Stung.
11-21-16.....	None	24 hrs.	+			
12- 6-16.....	7 or 8	12 min.	+ though the specimen had much food.			
12-10-16.....	2	20 min.	+ though well fed.			
12-11-16.....	6	3 min.	+	-	-	-
1-10-17.....	None	24 hrs.	+	-	-	-
1-18-17.....	2	2 min.	-	-	+	-
1-20-17.....	6-10	2 days	+	-	-	-
1-23-17.....	8	5 days	+	-	-	-
1-24-17.....	5	2 min.	-	-	-	+
1-26-17.....	1 or 2	2 min.	-	-	-	+
2-12-17.....	1	2 min.	-	-	-	+
4-19-17.....	10	2 min.	-	-	-	+ ¹
7-23-17.....	Few	2 min.	-	+	-	+
7-24-17a.....	6	10 min.	+	-	-	-
7-24-17b.....	None	10 min.	+	-	-	-
8-18-17b.....	None	5 min.	+	-	-	-
9-18-17d.....	1	24 hrs.	+	-	-	-
9-18-17e.....	Few	24 hrs.	-	+	+	-
9-18-17f.....	2	24 hrs.	+	-	-	-
9-18-17g.....	7	24 hrs.	+	-	-	-
9-18-17h.....	Few	5 min.	+	-	-	-
9-19-17a.....	Few	24 hrs.	+	-	-	-
9-19-17b.....	1	24 hrs.	+	-	-	-
9-19-17c.....	None	24 hrs.	+	-	-	-
9-19-17d.....	5	24 hrs.	+	-	-	-
9-19-17e.....	5	24 hrs.	+	-	-	-
9-19-17f.....	None	1 hr.	+	-	-	-
9-19-17g.....	7	1 hr.	+	-	-	-
9-19-17h.....	None	3 min.	+	-	-	-
10- 5-17a.....	None	6 hrs.	+	-	-	-
10-17-17a.....	1	24 hrs.	+	-	-	-
10-17-17c.....	6-8	24 hrs.	+	-	-	-
10-17-17e.....	None	24 hrs.	-	+	-	-
10-17-17f.....	None	24 hrs.	+	-	-	-
10-17-17h.....	8-10	24 hrs.	+	-	-	-
10-17-17j.....	None	24 hrs.	-	+	-	-
10-17-17l.....	Few	24 hrs.	-	+	-	-
10-17-17m.....	Few	24 hrs.	-	-	+	-
9-18-17e.....	Few	2 days	+	-	-	-
5- 3-23.....	Few	4 days	-	+	-	-
5- 3-23.....	10	24 hrs.	+	-	-	-
5- 8-23.....	None	24 hrs.	+	-	-	-

days with *Hydra* before they died without accepting any part of *Hydra*. Specimen 16 is the most significant one of the table. It was learned by observation that *Microstoma* could live in a small vessel of spring water for about nine days without food. Many died in less time under these conditions but none passed

¹ Rather by gastric juices than by nematocysts.

TABLE II.
NEMATOCYSTS OF *Microstoma* MANY.

Specimen.	Loaded with Nematocysts.	Without Food.	With <i>Hydra</i> .	<i>Hydra</i> Eaten.	<i>Hydra</i> Not Eaten.	Nematocysts Thrown from Enteron of <i>Microstoma</i> .
(1) 9-18-17B	1/2	1 day	2 days	+	-	-
(2) 9-18-17D	1/2	"	"	-	+	Sick; see notes.
(3) 9-18-17F	4/5	"	"	+	-	-
(4) 9-18-17G	1/2	"	"	+	-	-
(5) 9-19-17B	1/3	"	"	+	-	-
(6) 9-19-17C	1/2	"	3 days	+	-	-
(7) 9-19-17D	1/3	"	1 day	+	-	-
(8) 9-19-17D	Loaded	"	"	-	+	-
(9) 9-19-17E	2/3	"	3 days	+	-	-
(10) 9-19-17F	1/4	"	"	-	+	- Sick; see notes.
(11) 9-19-17G	1/4	"	"	+	-	-
(12) 9-19-17H	Loaded	"	"	-	+	-
(13) 9-19-17H	4/5	"	2 days	+	-	+
(14) 9-19-17A	1/2	"	"	+	-	-
(15) 9-19-17A	Loaded	No ¹	"	+	-	+
(16) 11-21-16	Loaded	9 days	"	-	+	Piece of tadpole liver accepted immediately after the <i>Microstoma</i> was taken from the <i>Hydra</i> of specimen (16).
(17) 5-6-23	1/2	4 days	4 days	-	+	
(18) 5-10-23	1/5	"	"	-	+	

the twelfth day. With this fact in mind a "loaded" specimen was kept away from food for nine days. It was placed with *Hydra* at the end of the ninth day and remained with it for two more days without accepting *Hydra*. This was not due to the *Microstoma's* condition having been so greatly lowered that it could not accept food. For when it was taken from *Hydra* and given a piece of tadpole liver, the *Microstoma* immediately accepted it. The facts tabulated in these two tables indicate that a *Microstoma* with few or no nematocysts attacks a *Hydra* much more readily than does one with many or a full quota.

Microstoma MAY EGEST CELLS OF *Hydra* AND RETAIN NEMATOCYSTS.

Another contrast may be drawn between the conduct of a *Microstoma* that has few or no nematocysts and that of one that has a full quota of nematocysts. If a *Microstoma* that contains a *Hydra*, so long ingested that the polyp has been reduced to a pulp, be placed under slight pressure it will discharge

¹ Not starved since it had eaten *Hydia* as specimen 14.

the green protoplasmic mass of the *Hydra's* body and retain the nematocysts within its enteron. This does not appear to be due to the fact that the pressure has held fast the mass of nematocysts; for, during the process of egesting the other *Hydra* material, the nematocysts are being thrust to and fro within the enteron. Under such condition a *Microstoma*, therefore, rejects the food and retains the nematocysts.

Microstoma MAY EGEST NEMATOCYSTS AND RETAIN CELLS
OF *Hydra*.

The senior author saw just the reverse of this. A *Microstoma* was loaded with nematocysts by being fed a *Hydra*. It was then starved until it would accept *Hydra*. While the *Microstoma* was kept under observation the *Hydra* was digested. When alimentation had been completed the indigestible nematocysts were thrown out of the mouth and rejected. When Dr. W. H. Taliaferro, now of Johns Hopkins University, was told this he expressed skepticism; his challenge was accepted by the senior author to have this demonstrated. From 10.58 A.M. to 11.14 A.M., September 19, 1917, Dr. Taliaferro kept the "loaded" *Microstoma* under observation. During this time he could see the nematocysts of a recently ingested *Hydra* within the enteron. At 11.14 A. M., he saw the nematocysts being discharged from the mouth of the "loaded" *Microstoma*.

From these observations it appears, therefore, that *Hydra* is eaten by *Microstoma* not primarily as a food but for its nematocysts. Thus, it is further suggested that the handling of the nematocysts by *Microstoma* is done in order that the rhabdo-coele may use these "stinging threads."

DISCHARGE OF NEMATOCYSTS BY MICROSTOMA A DOUBLE
PROCESS.

When one teases a *Microstoma*, that is armed with *Hydra's* nematocyst, the nematocyst in the immediate neighborhood of the stimulus oscillates to and fro at right angles to the surface of the body. If the teasing is maintained long enough, the nematocyst will be discharged at the object with which the *Microstoma*

was being touched. An observer cannot watch this double reaction on the part of the *Microstoma* when being stroked with a needle point without feeling that the first phase of reaction, viz., the oscillation of the nematocyst, is a threat and that the second phase of the reaction, viz., the discharge of the nematocyst, was an effort to use the nematocyst against the annoying object.

Microstoma USES ITS NEMATOCYSTS.

Finally we have been able to see *Microstoma* actually using the nematocysts that it had appropriated from *Hydra*.

October 13, 1917, the senior author placed some *Stenostomas* in a hanging drop with a *Microstoma* that had nematocysts. One of the *Stenostomas* was at once stung along its side. The wound caused the *Stenostoma's* body to rupture. After the *Microstoma* had thus wounded the victim, it at once swallowed it. Mr. Conway Zirkle saw this same *Microstoma* strike another *Stenostoma* in such fashion as to cause it to bend its body near the middle at right angles and to remain quiet in this contour until the *Microstoma* fell to ingesting it. In neither of these observations were the actual nematocysts seen entering or having entered the *Stenostoma*.

But on September 9, 1917, the senior author placed a large *Microstoma*, containing nematocysts, in a hanging drop of water with three very large dividing *Stenostomas*. The head of one of these *Stenostomas* came in contact with the right side of the *Microstoma's* head. Forthwith the *Stenostoma* contracted violently. From a wound at the tip of the body the mesenchyme oozed. As this material ran out of the body, it dragged with it two nematocysts that had their filaments and barbs ejected. The *Stenostoma's* body also developed a small blister to the left of the ruptured region of the epidermis. The *Microstoma* left the *Stenostoma*, which no longer moved from place to place, and later came back and attempted to ingest the *Stenostoma* as it lay struggling as though suffering from its wound.

September 19, 1917, the senior author placed a *Microstoma*, that had nematocysts, in a hanging drop of water with a single dividing *Stenostoma*. Eight times these two animals collided.

Sometimes these collisions caused the one animal to glide down along the side of the other; at other times the collisions would be head on. In each case the collisions were so evident as to cause a shunting of one or the other specimen. At 11.30 A.M. while *Microstoma* lay quiet, *Stenostoma* made a contact with *Microstoma* and stroked the entire left side of the *Microstoma* with its right ciliated pit. When *Stenostoma's* head had come to be in contact with the posterior end of *Microstoma*, the latter discharged two nematocysts into the region of *Stenostoma's* right ciliated pit. At once a rupture took place in the *Stenostoma's* epidermis and from this wound granular material oozed. Despite this breaking of the *Stenostoma's* body in the region in which it had been stung, the *Stenostoma* was anchored to *Microstoma* by the nematocysts, the poison sacs of which were yet held within the *Microstoma*. The two animals were anchored thus long enough for me to call Dr. I. F. Lewis to my side and make a demonstration of the situation to him. Soon after this demonstration was made, the two poison sacs left the *Microstoma's* body and the latter swam away. The *Stenostoma*, which up to the time it was stung had been incessantly active, lay quite inactive for ten minutes with two nematocysts hanging from the wound on the right side of its anterior end. After that it gradually recovered itself.

It has thus been demonstrated that *Microstoma* uses the nematocysts which it takes from the *Hydra*.

SUMMARY.

1. *Microstoma* manipulates the nematocysts of *Hydra* either (a) as a means of eliminating indigestible parts of its food, or (b) a means of defending itself.
2. The fact that the nematocysts are distributed uniformly over the surface of *Microstoma* within 12 hours after a *Hydra* has been ingested, suggests that this is not a process of elimination, (a), but is done in order that the nematocysts may be used, (b).
3. The fact that the nematocysts are retained indefinitely suggests that they have been taken up and retained for use.

4. The fact that this method of handling the nematocysts demands considerable work on the part of the endodermal cells and, in each case of a discharged nematocyst, the death of a cnidophage or mesodermal cell further suggests that it is all done with reference to use.

5. *Microstoma* incurs danger in attacking *Hydra*; for it is frequently stung and sometimes killed by the polyp. This too suggests that the *Microstoma* attacks *Hydra* for some peculiar end.

6. The *Microstoma* is able to draw nematocysts with which *Hydra* has stung it from its body, without suffering a rupture as does *Stenostoma*.

7. *Microstoma* with a few nematocysts attacks *Hydra* within a little time.

8. *Microstoma* having many nematocysts does not readily attack *Hydra*. The contrast between 7 and 8 suggests that *Microstoma* attacks *Hydra* not for food; but for its nematocysts.

9. The alimentary canal of a recently fed *Microstoma*, that has few nematocysts, may egest the "flesh" of *Hydra* and retain its nematocysts.

10. The alimentary canal of a *Microstoma* that has many nematocysts and is hungry, may egest the nematocysts of *Hydra* and retain the "flesh" of the polyp. The contrast between 9 and 10 suggests that *Microstoma* seeks primarily the nematocysts of *Hydra*.

11. The discharge of the nematocysts is a double process, involving (1) an oscillation to and fro of the nematocyst within its cnidophage, and (2) the actual discharge. The first phase may be carried on without being followed by the second phase. Neither of these has ever been seen except when some active body, like the experimenter's needle or an animal has brushed along the surface of *Microstoma*. This, too, suggests that the nematocysts are for use.

12. Finally *Microstoma* actually stings and paralyses other animals with the nematocysts it has appropriated from *Hydras* that it has eaten.

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EXPLANATION OF FIGURES.

PLATE I.

FIG. 1. A *Microstoma* slowly moving amid the bases of the tentacles of *Hydra*. $\times 50$.

FIG. 2. In response to the presence of *Microstoma* the *Hydra* has spread its expanding peristome along ventral surface of *Microstoma*. $\times 50$.

FIG. 3. The expanding peristome has glided posteriorly and dorsally, in direction indicated by arrow in Fig. 2, over surface of *Microstoma* as the *Hydra* flexed its body and pressed the *Microstoma* between its greatly expanded peristome and its bent body. From this embrace the *Microstoma* escaped. $\times 50$.

FIG. 4. Soon after this escape the *Microstoma* returned to be captured by the tentacles of *Hydra*. After the inception of ingestion the tentacles of *Hydra* were bent back so as to lie almost parallel to the axis of the polyp. The ingestion was a slow process. $\times 50$.

FIG. 5. Shows the manner in which a *Microstoma* was held by a *Hydra*. The *Microstoma* slowly glided out of this embrace between the ends of the tentacles. $\times 50$.

FIG. 6. When the *Microstoma* had escaped it showed two wounded regions—one bearing two nematocysts and the other three nematocysts. These nematocysts were at first embedded so that only their poison-sacs projected from the surface of the *Microstoma*. Very early a mass of mucus formed in each wounded region. The lateral mass moved posteriorly (as shown by arrow) and fused with the posterior one. As these masses of mucus grew in length, they dragged the five stinging threads from the body of the *Microstoma*. The combined mass of mucus, containing five nematocysts, eventually was cast off. After this the *Microstoma* moved about normally. $\times 50$.

FIG. 7. Shows a dividing *Microstoma* with a maximum charge of nematocysts at its surface. When the nematocysts are this frequent the *Microstoma* is said to be "loaded." $\times 50$.

